

Edge Agentic Systems: Retraining Approaches

Reference links on the data flywheel, self-training with confidence-based filtering, and self-supervised learning — compiled October 03, 2026

1. Data Flywheel (SLM-first agent design, LLM-to-SLM conversion)

Small Language Models are the Future of Agentic AI

<https://arxiv.org/html/2506.02153v1>

arXiv preprint — NVIDIA Research

The core position paper. Covers the SLM-first agent design and the LLM-to-SLM conversion algorithm built around usage-data collection, clustering, and fine-tuning.

NVIDIA Says Small Language Models Are The Future of Agentic AI

<https://cobusgreyling.substack.com/p/nvidia-says-small-language-models>

Substack — commentary

Readable summary focused on the data flywheel idea: examine usage data, cluster it by task, fine-tune task-specific small models.

Are Small Language Models the Future of Agentic AI?

<https://www.getmaxim.ai/blog/are-small-language-models-the-future-of-agentic-ai/>

Blog — commentary

Explains the paper's SLM/LLM definitions and the cost/efficiency argument for agentic workflows.

Small Language Models Are the Future of Agentic AI — NVIDIA Research

<https://medium.com/@aneessaheba.guddi/small-language-models-are-the-future-of-agentic-ai-nvidia-research-79177beaa069>

Medium — commentary

Short explainer on why small models suit repetitive agent steps and when to escalate to a large model.

2. Self-Training with Confidence-Based Filtering

Self-Training Large Language Models with Confident Reasoning

<https://arxiv.org/pdf/2505.17454>

arXiv preprint

LLMs fine-tuned on their own outputs, keeping only the ones the model rates as high-confidence.

Self-Training for End-to-End Speech Recognition

<https://arxiv.org/pdf/1909.09116>

arXiv preprint

Self-training with pseudo-label filtering and an ensemble; reports a 33.9% relative improvement in the noisy-speech setting.

Improving Cross-Lingual Reading Comprehension with Self-Training

<https://arxiv.org/pdf/2105.03627>

arXiv preprint

Includes a clear pseudo-label generation algorithm: keep an example only if its top confidence passes a threshold.

A Novel Self-training Approach for Low-resource Speech Recognition

<https://arxiv.org/pdf/2308.05269>

arXiv preprint

Starts with a strict confidence threshold and relaxes it each iteration, retraining until results stop improving.

Better Self-training for Image Classification through Self-supervision

<https://arxiv.org/pdf/2109.00778>

arXiv preprint

Explains how self-training differs from self-supervised learning, then combines the two.

3. Self-Supervised Learning

What is self-supervised learning? (DataCamp)

<https://www.datacamp.com/tutorial/self-supervised-learning>

Tutorial

Beginner-friendly comparison of self-supervised learning with supervised, unsupervised, and semi-supervised learning.

What is self-supervised learning? (IBM)

<https://www.ibm.com/think/topics/self-supervised-learning>

Overview

Covers pretext and downstream tasks and how self-supervised learning relates to unsupervised learning.

Self-Supervised Representation Learning: Introduction, Advances and Challenges

<https://arxiv.org/pdf/2110.09327>

arXiv survey

Survey covering pretext tasks, the pretraining-to-downstream workflow, and open challenges.

Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey

<https://arxiv.org/pdf/1902.06162>

arXiv survey

Defines pretext tasks, pseudo-labels, and downstream tasks, with a catalog of vision pretext methods.

Semi-Supervised and Unsupervised Deep Visual Learning: A Survey

<https://arxiv.org/pdf/2208.11296>

arXiv survey

Section 2.2.5 contrasts self-supervision (task-agnostic, no labels needed) with self-training.

Note: most entries above are arXiv preprints (directional, not yet peer-reviewed). Blog and tutorial links are for orientation, not citation.